

Xinran Liang

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EDUCATION

• Princeton University

Ph.D. in Computer Science

Sep 2022 - Present

Princeton, NJ

- Selected Coursework: Advanced Computer Vision, Probabilistic Inference in Reinforcement Learning, Theoretical Machine Learning, Systems and Machine Learning

• University of California, Berkeley

B.A. in Applied Mathematics and Data Science

Aug 2018 - May 2022

Berkeley, CA

- GPA: 4.0/4.0. Highest Distinction in General Scholarship (Top 3%).
- Selected Coursework: Data Structures, Efficient Algorithms, Artificial Intelligence, Machine Learning, Probability and Stochastic Processes, Optimization Models in Engineering, Deep Reinforcement Learning, Deep Unsupervised Learning, Deep Learning for Computer Vision

EXPERIENCE

• Princeton University

Graduate Student Researcher

Jan 2023 - Present

Princeton, NJ

- Post-training of VLM agents for long-horizon decision-making in video games: developed a unified training pipeline from SFT to RL that eliminates the need of human expert annotations, achieving competitive performance with frontier models.
- Proposed a method controlling pre-trained diffusion models for data augmentation through personalized fine-tuning, addressing limitations in vision datasets and improving model generalization capability.

• Intuit AI Research

Research Intern

Jun 2025 - Aug 2025

Mountain View, CA

- A two-stage approach for risk monitoring in multi-turn interaction systems, including a multi-instance learning model for risk scoring and a single preference-conditioned triggering policy trained via RL.

• Berkeley Artificial Intelligence Research

Undergraduate Student Researcher

Jan 2021 - Sep 2022

Berkeley, CA

- A simple and efficient exploration method based on epistemic uncertainty from human feedback, and improved preference-based reinforcement learning algorithms in complex decision making tasks.
- Developed an embodied learning framework that leverages active exploration for visual representation learning and perception tasks in complex 3D environments; proposed incorporating action supervision from agent movements to enhance visual representation learning.

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, W=WORKSHOP, P=PREPRINT

- [P.2] **Odysseus: Scaling VLMs to 100+ Turn Decision-Making in Games via Reinforcement Learning**
Chengshuai Shi*, Wenzhe Li*, **Xinran Liang***, Yizhou Lu, Wenjia Yang, Ruirong Feng, Seth Karten, Ziran Yang, Zihan Ding, Gabriel Sarch, Danqi Chen, Karthik Narasimhan, Chi Jin
In *arXiv* 2026.
- [W.2] **Personalized Generative Models for Contextual Debiasing**
Xinran Liang, Esin Tureci, Prachi Sinha, Ye Zhu, Vikram Ramaswamy, Olga Russakovsky
In *CVPR 2026 Workshop on Synthetic Data for Computer Vision and Generative Models for Computer Vision*.
- [W.1] **Sparse Evidence, Controllable Triggers: Weakly Supervised Early Failure Alerting in Multi-Turn LLM Interactions**
Avinash Baidya, **Xinran Liang**, Ruocheng Guo, Kamalika Das, Xiang Gao
Under Review at EMNLP 2026. In *NeurIPS 2025 Workshop on Multi-Turn Interactions in LLMs*.
- [P.1] **ALP: Action-Aware Embodied Learning for Perception**
Xinran Liang, Anthony Han, Wilson Yan, Aditi Raghunathan, Pieter Abbeel
In *arXiv* 2023.
- [C.1] **Reward Uncertainty for Exploration in Preference-based Reinforcement Learning**
Xinran Liang, Katherine Shu, Kimin Lee, Pieter Abbeel
In *ICLR* 2022.

SKILLS

- **Language/Framework:** Python, PyTorch, Mujoco, vLLM, sgLang, LLaMA-Factory, veRL
- **Machine Learning:** Vision Language Models, Large Language Models, Reinforcement Learning, Generative Models

HONORS AND AWARDS

- **Department Citation in Data Science** (UC Berkeley) *May 2022*
- **Outstanding Graduate Student Instructor Award** (UC Berkeley) *May 2022*

TEACHING EXPERIENCE

- **Princeton University** *Sep 2023 - May 2024*
Graduate Student Instructor
 - Introduction to Machine Learning (Fall 2023). Algorithms and Data Structures (Spring 2024).
- **University of California, Berkeley** *Jan 2020 - May 2022*
Undergraduate Student Instructor
 - Probability for Data Science (Spring 2020 - Spring 2022). Principles and Techniques of Data Science (Summer 2020).

ACADEMIC SERVICE

- **Reviewer:** ECCV 2024, ICLR 2025-2026, CVPR 2025-2026, NeurIPS 2026, BMVC 2026, CVPR Workshop on Explainable AI for Computer Vision, NeurIPS Workshop on Multi-Turn Interactions in Large Language Models